Journal of Power Sources 603 (2024) 234443

Contents lists available at ScienceDirect

# Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour

# Highly conductive and transparent metal microfiber networks as front electrodes of flexible thin-film photovoltaics

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## HIGHLIGHTS

- Metal microfiber (MF) transparent conductors (TCs) were fabricated.
- Ni and Cu MF TCs were formed by electrospinning and electroplating processes.
- MFs show exceptionally low electrical resistivity with high transmittance.
- $Cu(In,Ga)Se_2$ solar cells with MFs outperformed traditional grids.
- Proposed MFs are expected to replace grids and ribbons in photovoltaic modules.

## ARTICLE INFO

*Keywords:*
Thin-film solar cells
Ni fibers
Cu fibers
Transparent electrodes
Gridless

## ABSTRACT

Simultaneously enhancing the optical transmittance and electrical conductivity of transparent conductors (TCs) applicable in various optoelectronic devices is a long-standing challenge. Herein, we present an innovative approach for fabricating electroplated Ni and Cu microfiber networks (NiMF and CuMF) as highly conductive TCs to realize high efficiency and desired aesthetics in thin-film solar cells and modules. The metal microfibers (MFs) are fabricated using electrospun polyacrylonitrile nanofibers as the template. The large cross-sectional aspect ratio of the metal MF networks remarkably and concurrently improves their electrical conductivity and optical transmittance. Between the NiMF and CuMF TCs, the CuMF sample exhibits a superior figure of merit owing to its exceptionally low electrical resistivity. The metal MF TC is a promising alternative to conventional patterned grids used in flexible $Cu(In,Ga)Se_2$ thin-film solar cells, because it effectively reduces the series resistance, which is advantageous for large-area cells. The CuMF can be successfully employed as a ribbon to maintain the solar cell performance in centimeter-scale cells connected in series. The outstanding performance of the metal MF TCs indicates their potential to eliminate complicated monolithic integration processes or front grids and ribbons in flexible thin-film solar cells and modules.

## 1. Introduction

Transparent conductors (TCs) are extensively applied as electrodes in devices that absorb or emit light, such as thin-film solar cells, displays, sensors, and electronic paper [1]. Numerous strategies have been explored to achieve a combination of superior optical transmittance and high electrical conductivity, and various materials, including transparent conducting oxides (TCOs), ultrathin metallic layers, and nanoscale conductive materials (e.g., silver nanowires (AgNWs), carbon nanotubes (CNTs), and graphene), have been utilized as optically

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https://doi.org/10.1016/j.jpowsour.2024.234443
Received 23 February 2024; Received in revised form 25 March 2024; Accepted 26 March 2024
Available online 4 April 2024

transmitting electrical conductors [2–7].

Cu(In,Ga)$Se_2$ (CIGS) thin-film solar cells have garnered attention as power sources for urban self-sufficiency owing to their high efficiency (>23%) and applicability in flexible building-integrated photovoltaic (BIPV) systems [8,9]. The TC electrodes of CIGS thin-film solar cells commonly incorporate thin films of TCOs, such as heteroatom-doped zinc oxide (ZnO) or indium tin oxide (ITO) [10,11]. The TCO deployed in CIGS thin-film solar cells requires superior characteristics, such as high transmittance and low resistance for achieving optimal light absorption and carrier transport efficiency, respectively. The higher than desired sheet resistance ($R_{sh}$, typically tens of Ω/□) of TCO films hinders the efficient transport of generated electrons solely within them. Consequently, the manufacturers of CIGS thin-film solar cells and modules typically utilize a monolithic integration process to reduce electron transport distances [12]. However, translating this process to flexible substrates presents significant challenges owing to requirements for high-quality insulation layers for metallic foils and the incompatibility of structuring processes with roll-to-roll processing [13,14]. Consequently, commercial flexible CIGS photovoltaics frequently incorporate front grids featuring busbar and finger patterns [15–17]. In highly efficient crystalline silicon solar modules, a multi-busbar metal grid is employed and millimeter-wide metal grids are selected for achieving optimal electrical conductance [18]. These broad grid patterns hinder light transmission and conspicuous, which limit their applicability in building-integrated photovoltaics and portable devices, where aesthetics is an important consideration. It is therefore imperative to significantly decrease the resistance of TCs to minimize or eliminate the grid area, which affects the optimization of the grid pattern geometries and cell dimensions [19,20].

Metal fibers offer a promising solution for achieving a notably elevated cross-sectional aspect ratio than those of metal grid films, allowing a high aperture ratio and reduced resistance. While nanoscale metal fibers or wires have been explored extensively, their restricted cross-sectional area and high surface area limit their conductivity and chemical stability, respectively [21–24]. Microscale fibers composed of corrosion-resistant materials have emerged as viable alternatives to conventional nanostructured networks for various applications such as batteries, fuel cells, supercapacitors, and solar cells [25–31]. Despite the development of highly transparent and conductive metal fibers in earlier studies [32–34], very few studies have investigated their utilization as front contacts in thin-film solar cells.

In this study, we produced highly conductive Ni and Cu microfiber (MF) networks as TCs using an electrospun polymer fiber template and investigated the performance of the MF TCs in CIGS thin-film solar cells. The objectives of the study were to enhance the efficiency of CIGS thin-film solar cells and eliminate patterned grids. Herein, we demonstrate the enhanced performance of CIGS photovoltaic devices incorporated with MF networks in single and serially connected configurations.

## 2. Experimental section

The metal MFs were produced through a series of processes illustrated in Fig. 1. Initially, an 8 wt% solution of polyacrylonitrile (PAN, Mw = 150 kDa, Sigma-Aldrich, USA) in *N,N*-dimethylformamide (DMF, 99.8%, Sigma-Aldrich, USA) was electrospun onto a flat collector for 10 s. The flow rate of the solution, sprayed through a needle (25 gauge, EFD, USA) using a syringe pump (Legato 100, KD Scientific, USA), was maintained at 200 μL/h. Electrospinning was performed at a voltage of 5 kV using a high-voltage DC supply (EL20P2; Glassman High Voltage, USA). Thereafter, the electrospun PAN nanofibers (NFs), measuring ~250 nm in diameter, were deposited onto a square copper frame (dimensions: 4.5 cm × 3 cm) and Pt seeds were sputtered onto them. Subsequently, Ni or Cu electroplating was performed on the Pt-seeded PAN NFs at a voltage of 6 V for different durations ($t_{EP}$). The $t_{EP}$ was found to correlate with the diameter of the fabricated metal MFs. The resulting MFs were rinsed with distilled water and thoroughly dried under $N_2$ for a few minutes. Finally, the Ni or Cu MFs were separated from the frame and transferred to a solar cell with the following architecture: ITO/i-ZnO/CdS/CIGS/Mo/stainless-steel. To enhance contact between the metal MFs and ITO surface, a piece of 56 μm thick commercial adhesive tape (Scotch® MagicTM Tape, 3 M, USA) was used to affix the MF network to the device. This adhesive tape, known as a transparent diffusive layer (TDL) [35], was chosen because of its high transparency and haze characteristics.

CIGS thin-film solar cells with the structural configuration of ITO/i-ZnO/CdS/CIGS/Mo/stainless steel were manufactured according to our previously documented baseline process [35–44]. The Mo back contact was applied to a stainless steel substrate through direct current sputtering. Subsequently, a CIGS absorption layer with a thickness of 2.2 μm was deposited via the independent thermal co-evaporation of Cu, In, Ga, and Se, employing a multi-stage evaporation sequence of In–Ga–Se, Cu–Se, and In–Ga–Se. A CdS buffer layer was deposited through a chemical bath deposition process. Then, a 70 nm thick i-ZnO thin film followed by a 150 nm thick ITO thin film were deposited by radio-frequency sputtering. Finally, the solar cell areas were defined by mechanical scribing for electrical isolation.

The morphologies of the metal MFs were examined by scanning electron microscopy (SEM; SU8230, Hitachi, Japan). Their optical transmittance was assessed using a UV–visible spectrophotometer (Lambda 950, PerkinElmer, USA) [45]. The sheet resistance ($R_{sheet}$) of the metal MF TC was measured using a four-point-probe measuring system (CMT-SERIES, Chang Min Co., Ltd., Korea). The current density–voltage (*J–V*) characteristics of the CIGS thin-film solar cells were determined using an *I–V* source meter (Keithley 2400, Tektronix, USA) under Air Mass 1.5 global spectrum conditions (irradiance: 1000 W $m^{-2}$) at room temperature.

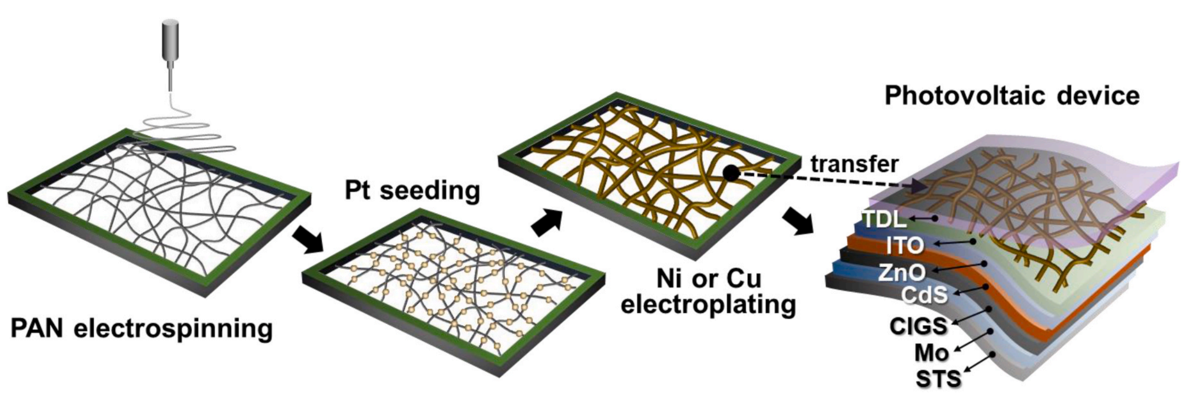


**Fig. 1.** Schematic of the fabrication of metal microfibers and their transfer onto CIGS thin-film solar cells. The process includes the electrospinning of polyacrylonitrile fibers, metal (Cu or Ni) electroplating, and metal microfiber transfer processes.

## 3. Results and discussion

Fig. 2 displays the photographs of the NiMF and CuMF samples, sized approximately 4.5 cm × 3 cm (the blue dotted line area), overlaid on printed paper. The two rectangular MF samples have a uniform hue over the entire area, as observed with the naked eye. We anticipated that the color of the metal MF film can be modulated by changing the metal species for achieving desired aesthetic appearance. The NiMFs (Fig. 2(a)) and CuMFs (Fig. 2(b)) appear silvery white and brownish orange, respectively, reflecting the natural colors of bulk Ni and Cu metals. Further, as the metal electroplating time ($t_{EP}$) increased, that is, as the fiber thickness increased, the MF network became more visible.

The SEM images in Fig. 3 clearly reveal that the MF diameter increased with increasing $t_{EP}$. As the $t_{EP}$ of Ni was increased to 20, 40, and 60 s, the diameter of the NiMF increased to 3.6, 3.9, and 4.2 μm, respectively. Similarly, for CuMFs, the fiber diameter increased to 2.3, 4.0, and 5.2 μm as the $t_{EP}$ was increased to 30, 40, and 50 s, respectively. The growth rates of Ni and Cu differed and were not directly proportional to the electroplating time. As the micron-sized metal fibers were grown on the surface of PAN NFs with a diameter of approximately 250 nm, the interconnections of the metal MFs were fused well, which minimized the contact resistance between individual fibers.

Fig. 4(a and b) shows the transmittances of the NiMF and CuMF films prepared under different condition along with their $R_{sheet}$ values. As the $t_{EP}$ increased, the transmittance of the metal MF film decreased over the entire wavelength range (400–1100 nm). Transmittance is inversely proportional to the area occupied by the MFs, that is, the MF film diameter. The average transmittance ($T_{avg}$) of the NiMF film in the 400–1100 nm range decreased with increasing $t_{EP}$ (it was 91.8, 87.8, and 81.6%, respectively, for the samples obtained with a $t_{EP}$ of 20, 40, and 60 s). For the CuMF film, as the $t_{EP}$ increased to 30, 40, and 50 s, the $T_{avg}$ decreased to 89.5, 89.0, and 83.7%, respectively. The $R_{sheet}$ decreased rapidly with increasing $t_{EP}$ for both the samples (NiMF and CuMF). We note that the difference in fiber diameter shown in Fig. 3 may not necessarily result in the arithmetic difference in transmittance shown in Fig. 4 due to differences in fiber sparsity that may be caused by time and flow rate variations in the electrospinning process.

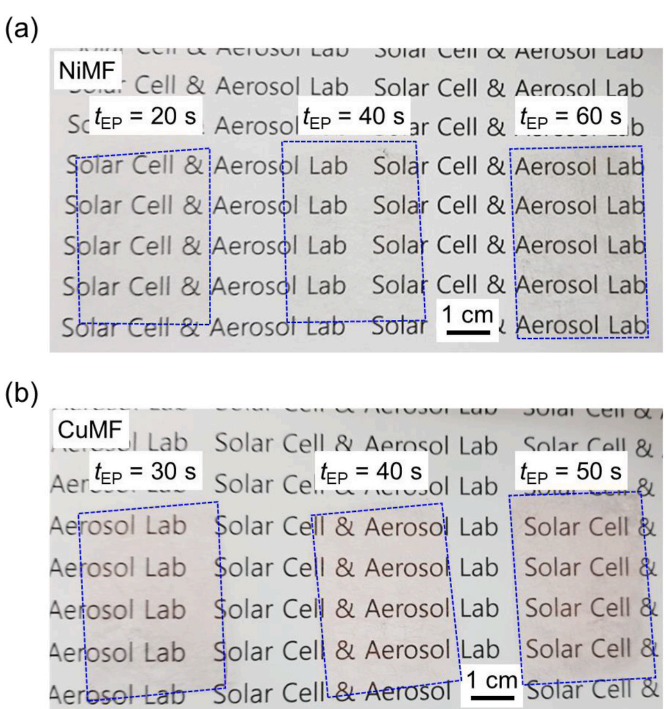


**Fig. 2.** Physical appearance of the fabricated (a) Ni MFs ($t_{EP}$ = 20, 40, and 60 s) and (b) Cu MFs ($t_{EP}$ = 30, 40, and 50 s).

The performance of the fabricated MFs as electrode substitute for solar cells was evaluated. According Equation (1), the optical transmittance ($T$) of a TC is in a trade-off relationship with its $R_{sheet}$:

$$T = \left[1 + \frac{Z_0}{2R_{sheet}}\left(\frac{\sigma_{DC}}{\sigma_{Opt}}\right)^{-1}\right]^{-2} = \left[1 + \frac{188.5}{R_{sheet}}\left(\frac{\sigma_{DC}}{\sigma_{Opt}}\right)^{-1}\right]^{-2}, \quad (1)$$

where $\sigma_{DC}$, $\sigma_{Opt}$, and $Z_0$ are the DC conductivity, optical conductivity, and impedance of free space, respectively [46]. The ratio of the DC conductivity to optical conductivity, denoted as $\sigma_{DC}/\sigma_{Opt}$, serves as a figure of merit (FOM) for evaluating TCs.

Fig. 4(c) presents the transmittance at 550 nm ($T_{550}$) as a function of $R_{sheet}$ for the NiMF and CuMF films (red and green symbols). The dashed curves represent the $\sigma_{DC}/\sigma_{Opt}$ values for different materials developed in previous studies. The $\sigma_{DC}/\sigma_{Opt}$ values of the NiMF and CuMF films are approximately 5000 and 27,000, respectively. The data in Fig. 4(c) reveal that the metal MF films fabricated in this study exhibit superior $\sigma_{DC}/\sigma_{Opt}$ values compared with those of the other reported TCs, including ITO films, AgNWs, patterned metal grids, and single-walled CNTs (SWCNTs) [4,46–50]. The CuMF film exhibited significantly higher $\sigma_{DC}/\sigma_{Opt}$ values than the NiMF film. This is because of the inherently higher electrical conductivity of Cu compared with that of Ni. Evidently, the CuMF film achieved an FOM rivalling that of a structure combining TCO and a patterned metal grid (~27,000). The produced NiMF and CuMF exhibit significantly lower sheet resistance compared to the cases of metallic (conducting) networks such as Cu, CuS, Au, Ag, SWCNT, and PEDOT:PSS applied in solar cells (shown in Table S1).

Fig. 5 presents representative photographs and schematics of the MF-applied solar cells. Fig. 5(a) shows a CIGS solar cell (area of 5.25 cm$^2$) with a NiMF film prepared with $t_{EP}$ = 60 s (denoted as NiMF-60s). Fig. 5(b) shows two CIGS thin-film solar cells connected in series. In this case, a CuMF film prepared with $t_{EP}$ = 50 s (CuMF-50s) was placed on the first (left) solar cell that was contacted with the bottom electrode of the second (right) solar cell; in addition, a CuMF prepared with $t_{EP}$ = 40 s (CuMF-40s) was applied to the top of the second (right) solar cell and contacted with the top electrode. After placing the MF on the solar cell, an adhesive TDL was affixed on the solar cell area, excluding the electrical contact area. The TDL was used to improve the contact between the MF film and solar cell surface and enhance the optical gain by facilitating the diffusion of incident light, which prolongs the optical path length in the absorber [35,51,52]. The cell area was isolated by mechanical scribing, and the bottom contact was formed by scratching to expose the Mo layer and then coating indium.

The current density–voltage ($J$–$V$) curves of the single solar cells using NiMF and CuMF TCs prepared with different $t_{EP}$ values are shown in Fig. 6(a). The $J$–$V$ curves of a solar cell without the MF film and another cell with a patterned Ni/Al grid are also shown for comparison. Fig. 6(b) shows the $J$–$V$ curves of single cells using CuMF-40s and CuMF-50s, and a cell connected in series. The CuMF-40s and CuMF-50s solar cells used for single-cell measurements were not the same as the ones used in the serially connected solar cells.

The performance parameters of the MF films and MF-applied solar cells are listed in Tables 1 and 2. In the cases of both NiMF- and CuMF-containing cells, the fill factor ($FF$) increased with increasing $t_{EP}$, because the $R_{sheet}$ decreased. Conversely, the $T_{avg}$ of the MF decreased with increasing $t_{EP}$, and the short-circuit current density ($J_{SC}$) changed negligibly. This is possibly because the optical loss was compensated by improved carrier collection owing to the increased electrical conductivity of the device, although the transmittance of the MF film decreased with increasing $t_{EP}$. Moreover, despite the absorption loss in the TDL/MF layer, the reason for the high $J_{SC}$ is attributed to the increased optical path length within the CIGS absorption layer due to the improved diffusive transmission by the TDL [51,52], as evidenced by the EQE results in Figure S1 (not identical to the solar cells discussed in the main text).

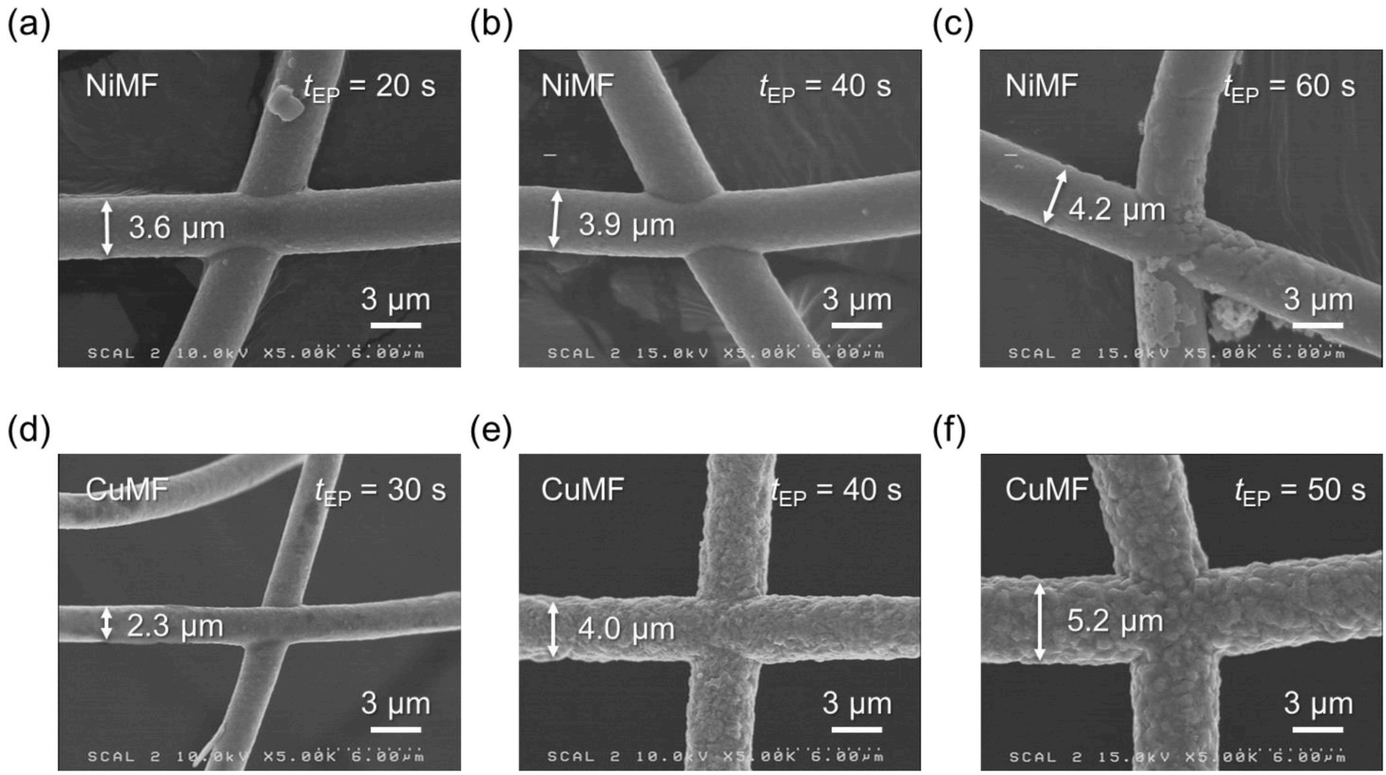


**Fig. 3.** SEM images of the areas of the TCs with interconnections: NiMFs ($t_{EP}$ = (a) 20 s, (b) 40 s, and (c) 60 s) and CuMFs ($t_{EP}$ = (d) 30 s, (e) 40 s, and (f) 50 s).

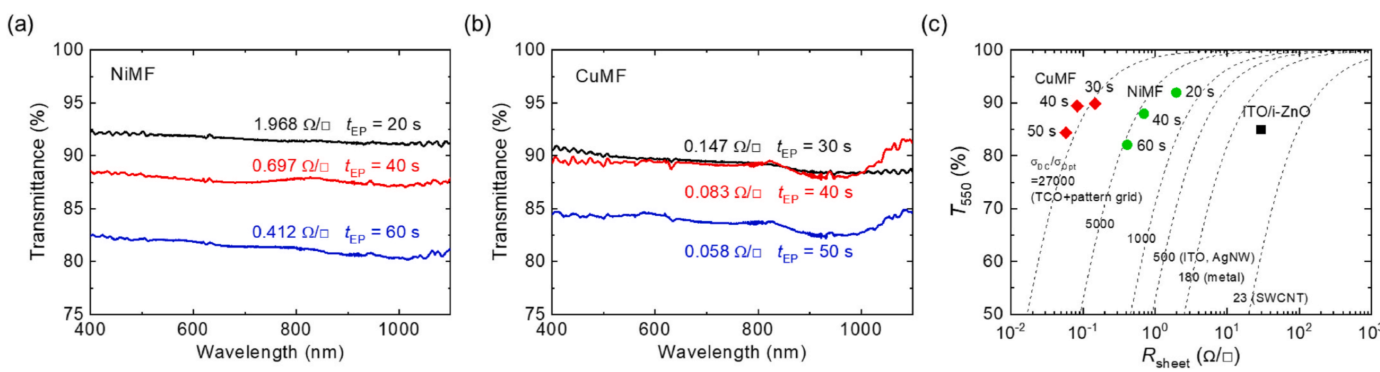


**Fig. 4.** Optical transmittances of the (a) NiMF and (b) CuMF TCs. The corresponding $R_{sheet}$ value is shown in each spectrum. (c) $T_{550}$ and $R_{sheet}$ values of the NiMF and CuMF TCs fabricated using various $t_{EP}$ values. The $T_{550}$ and $R_{sheet}$ values of the ITO/i-ZnO bilayer (window layer) of the CIGS solar cell are also plotted for comparison. Dashed lines show representative $\sigma_{DC}/\sigma_{Opt}$ values from previous reports for pattern grid-added TCO, ITO, AgNW, metal, and SWCNT samples [4,46–50].

When only the indium top contact was formed, instead of using an MF film or grid (Figure S2(a)), the *FF* and $J_{SC}$ were very low. This is because the travel length of the generated carriers to the top contact was excessively long, resulting in significantly limited carrier collection. The *FF* values of the solar cells employing the CuMF TC were higher than those of the cells using a patterned Ni/Al grid (shading area of 3.2%). The *FF* value of 67.2% in CuMF-50s condition was comparable to that of 70.5% in a small area cell (effective area of 0.47 cm$^2$) as shown in Figure S3. For cells using the NiMF TC, the *FF* of the cell with NiMF-40s was comparable to that of the cell using the grid, whereas the FF of the cell with the NiMF-60s increased by 11.2%p. The detailed dimensions of the patterned grid and solar cell area are shown in Figure S2(b).

Additional processes, including the deposition of an insulating layer, are required to manufacture solar modules using conducting materials, such as stainless steel substrates, because monolithic integration is not possible. Therefore, to utilize the excellent electrical conductivity of the CuMF TC and to confirm its potential not only as a single-cell electrode but also as a ribbon for connecting cells, two CuMF-applied solar cells were fabricated on a flexible stainless steel substrate, and the cell performance was evaluated. As shown in Fig. 5(b) and Table 2, the CuMF-50s-employed cell with an efficiency of 13.23% was connected to the bottom contact of another cell with an efficiency of 13.10% using CuMF-40s. Upon comparing the measured performance parameters with the theoretical values after the series integration (INT) of two cells (A (CuMF-40s) and B (CuMF-50s)), the efficiency change ($(\Delta\eta = \Delta\eta_{INT} - (\eta_A + \eta_B)/2)$) was −1.04%p (−7.86%), $V_{OC}$ change ($\Delta V_{OC} = \Delta V_{OC,INT} - (V_{OC,A} + V_{OC,B})$) was −19 mV (−1.38%), $J_{SC}$ change ($\Delta J_{SC} = \Delta J_{SC,INT} - ((J_{SC,A} \times 4.125 + J_{SC,B} \times 5.25)/9.375/2)$) was −0.22 mA/cm$^2$ (−1.49%), *FF* change ($\Delta FF = \Delta FF_{INT} - (FF_A + FF_B)/2$) was −3.25%p (−5.01%), and $P_{max}$ change ($\Delta P_{max} = \Delta P_{max,INT} - (P_{max,A} + P_{max,B})$) was −9.75 mW (−7.89%). The $V_{OC}$ and $J_{SC}$ losses were small, and the decrease in the efficiency and $P_{max}$ were due to *FF* degradation. For using the metal MF as a ribbon for connecting the cells, the *FF* loss may be minimized by maximizing the conductivity of the MF by increasing its

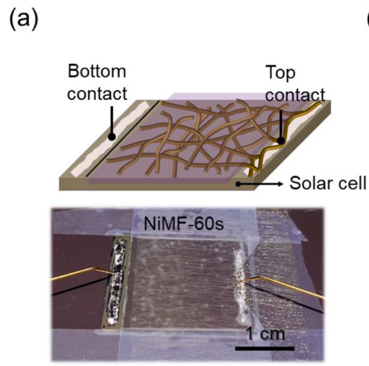


**Fig. 5.** Schematic and real images of the exemplary metal microfiber-applied CIGS thin-film solar cells: (a) NiMF ($t_{EP}$ = 60 s)-based single solar cell with an effective area of 5.25 cm$^2$, and (b) two serially connected solar cells using CuMFs ($t_{EP}$ = 40 and 50 s) with an effective area of 9.375 cm$^2$. The connection area represents the area where the CuMF of the first (left) cell and the bottom contact point of the second (right) cell contact.

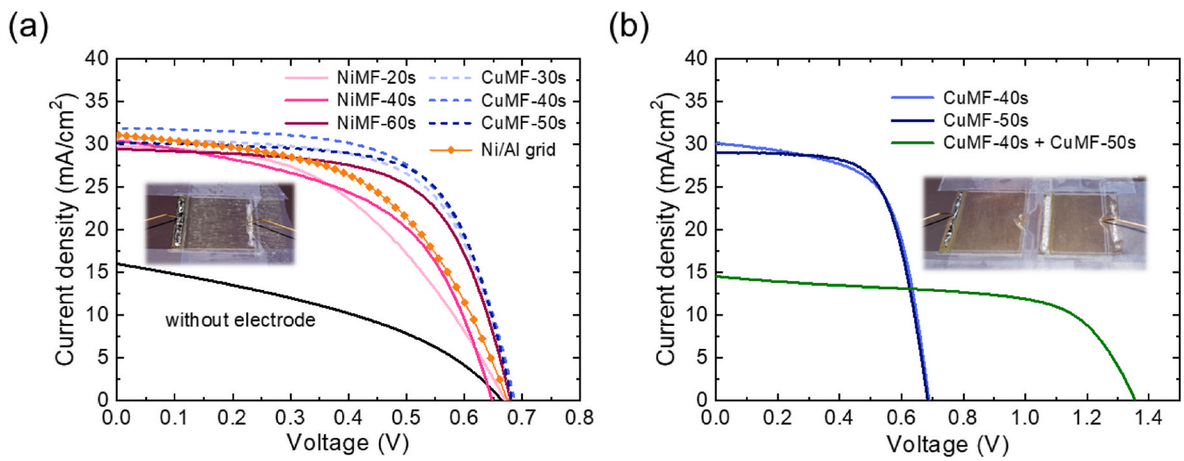


**Fig. 6.** (a) *J*–*V* curves of the NiMF- and CuMF-applied single CIGS thin-film solar cells with an effective area of 5.25 cm$^2$. The *J*–*V* curves of the solar cells with a patterned Ni/Al grid (orange line) and without any grid electrode (black line) are also plotted for comparison. (b) *J*–*V* curve of a serially connected solar cells using two different CuMFs ($t_{EP}$ = 40 and 50 s); the *J*–*V* curve of each sample before the connection is also shown. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

**Table 1**
$R_{sheet}$ and $T_{avg}$ values of the NiMF and CuMF TCs produced using different electroplating times ($t_{EP}$), and the corresponding solar cell performance parameters. The performance parameters of solar cells with a patterned Ni/Al grid and without any grid electrode are also shown for comparison.

| Metal fiber property | | | | Solar cell performance | | | | |
|---|---|---|---|---|---|---|---|---|
| Material | $t_{EP}$ (s) | $R_{sheet}$ (Ω/□) | $T_{avg}$ (%) | $\eta$ (%) | $V_{OC}$ (V) | $J_{SC}$ (mA/cm$^2$) | *FF* (%) | Effective Area (cm$^2$) |
| NiMF | 20 | 1.968 | 91.8 | 9.50 | 0.671 | 29.85 | 47.4 | 5.25 |
| | 40 | 0.697 | 87.8 | 10.29 | 0.648 | 30.47 | 52.1 | |
| | 60 | 0.412 | 81.6 | 12.69 | 0.681 | 29.44 | 63.3 | |
| CuMF | 30 | 0.147 | 89.5 | 13.36 | 0.683 | 30.65 | 63.9 | 5.25 |
| | 40 | 0.083 | 89.0 | 13.88 | 0.688 | 31.93 | 63.2 | |
| | 50 | 0.058 | 83.7 | 13.85 | 0.683 | 30.18 | 67.2 | |
| Without electrode | | – | – | 4.12 | 0.665 | 16.00 | 38.7 | 5.25 |
| Ni/Al grid | | – | – | 10.78 | 0.674 | 30.54 | 52.3 | 4.75 |

**Table 2**
$R_{sheet}$ and $T_{avg}$ values of the CuMF TCs fabricated with electroplating times ($t_{EP}$) of 40 and 50 s, and the photovoltaic performance parameters of the solar cells serially connected using the CuMF TC.

| Metal fiber property | | | | Solar cell performance | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Material | $t_{EP}$ (s) | $R_{sheet}$ (Ω/□) | $T_{avg}$ (%) | $\eta$ (%) | $V_{OC}$ (V) | $J_{SC}$ (mA/cm$^2$) | *FF* (%) | $P_{max}$ (mW) | Effective Area (cm$^2$) |
| CuMF | 40 | 0.126 | 89.0 | 13.10 | 0.689 | 30.12 | 63.2 | 54.03 | 4.125 |
| | 50 | 0.037 | 83.7 | 13.23 | 0.685 | 29.05 | 66.5 | 69.47 | 5.25 |
| | Integrated | – | – | 12.13 | 1.355 | 14.54 | 61.6 | 113.75 | 9.375 |

thickness or density. In addition, the cell-to-module loss can possibly be reduced by optimizing the cell placement and spacing between the cells during module manufacturing.

The time-dependent conductivity degradation of the metal MFs was negligible. To evaluate their stability, the as-fabricated bare MFs were stored in a dry atmosphere at room temperature for approximately nine months and their characteristics were evaluated again. As shown in Table S2, the $R_{sheet}$ change rate ranged from −2.18% to +7.75% for NiMFs and from −3.45% to +7.23% for CuMFs. Both the metal MFs maintained excellent conductivity over a long duration. Likewise, there was no change in optical transmittances of the metal MFs.

## 4. Conclusions

NiMFs and CuMFs were produced using an easily scalable and facile electroplating technique for use as TCs. The so-produced metal MFs exhibited significantly improved electrical conductivities and thus a higher FOM compared to traditional TCs. The CuMF network exhibited a superior FOM to that of the NiMF network owing to its exceptionally low resistivity. The well-fused interconnections of the randomly distributed MFs substantially decreased the resistance of the MF network, providing pathways for carrier collection in photovoltaic devices. Flexible CIGS thin-film solar cells employing the MFs instead of patterned grids exhibited excellent photovoltaic performance. Increasing the thickness of the MFs (using a higher electroplating time) improved the conductivity, leading to a larger *FF*, while causing negligible degradation in $J_{SC}$. The high conductivity of the CuMF facilitated the series connection of centimeter-scale solar cells without significant loss in device performance. According to the study findings, metal MFs have the potential to replace conventional front-patterned grids and ribbons in thin-film photovoltaic modules, and can facilitate improved efficiency, flexibility, processability, and aesthetic appeal in BIPV systems.

## CRediT authorship contribution statement

**Dae-Hyung Cho:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Woo-Jung Lee:** Writing – review & editing, Methodology, Data curation. **Tae-Ha Hwang:** Writing – review & editing, Methodology. **Jung-woo Huh:** Writing – review & editing, Methodology. **Sam S. Yoon:** Writing – review & editing, Supervision. **Yong-Duck Chung:** Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Acknowledgments

This work was supported by the program of Phased development of carbon neutral technologies (2022M3J1A1063019) through the National Research Foundation (NRF) funded by the Ministry of Science, ICT, and by an Electronics and Telecommunications Research Institute (ETRI) grant funded by the Korean government (24YB1500).

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2024.234443.

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